Lab 8

MSDS684_S70_Reinforcement Learning

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May 2026

Project Overview

This project investigates the trade-offs between different reinforcement learning paradigms by comparing three fundamentally distinct algorithms on the CartPole-v1 environment: Q-learning (off-policy TD control), SARSA (on-policy TD control), and REINFORCE with baseline (policy gradient method).

According to Reinforcement Learning: An Introduction, these algorithms represent three major families of reinforcement learning approaches. Q-learning and SARSA (Chapter 6) rely on bootstrapping and value function estimation, while REINFORCE (Chapter 13) directly optimizes a parameterized policy using gradient ascent. This distinction leads to different behaviors in terms of sample efficiency, stability, and convergence.

The CartPole-v1 environment consists of a continuous state space (cart position, velocity, pole angle, angular velocity) and a discrete action space of two actions (left or right force). The reward structure is simple: the agent receives +1 for every time step the pole remains upright, and episodes terminate when the pole falls or a time limit is reached (500 steps). The goal is to maximize episode length.

Because Q-learning and SARSA are tabular methods, the continuous state space must be discretized. This introduces approximation errors and limits generalization across states. In contrast, REINFORCE uses neural networks, allowing it to generalize across continuous space but introducing instability due to high-variance gradient estimates.

Hypothesis:

REINFORCE will achieve faster initial learning due to function approximation and direct policy optimization, but will exhibit higher variance and instability. Q-learning and SARSA will learn more slowly due to discretization and limited generalization, but will converge more stably. Between the two, Q-learning is expected to converge slightly faster due to its off-policy update using the max operator, while SARSA will be more conservative due to incorporating exploration into its updates.

This hypothesis is grounded in the update mechanisms:

- Q-learning: uses $\max_a Q(s', a)$, promoting greedy improvement
- SARSA: uses $Q(s', a')$, reflecting exploratory behavior
- REINFORCE: updates via $\nabla \log \pi(a \mid s) \cdot G$, which has inherently high variance

Deliverables

GitHub Repository:

https://github.com/MPalacios-Analytics/DataScience_Projects/tree/68fce93f0cc0ecabc612f256e96168145a7c0fa8/06_Reinforcement_Learning/Lab_8

Implementation Summary

Three algorithms were implemented and compared on CartPole-v1 over 600 episodes and 3 random seeds per algorithm.

Q-learning and SARSA use tabular Q-functions with discretization (6 bins per state dimension). Both use ε-greedy exploration ($\varepsilon = 0.1$), learning rate $\alpha = 0.1$, and discount factor $\gamma = 0.99$. These values were chosen to balance exploration and convergence: $\varepsilon = 0.1$ ensures continued exploration but imposes a performance ceiling, while $\alpha = 0.1$ provides reasonably fast updates without excessive instability.

However, the choice of only 6 bins per dimension results in a coarse state representation, which significantly limits performance in a continuous environment like CartPole.

REINFORCE with baseline uses two neural networks (policy and value networks with 32 hidden units). A learning rate of 0.01 was used for both networks. The baseline reduces variance by estimating state values, while return normalization further stabilizes updates.

All experiments were run with seeds [0, 42, 123], and results are reported as mean ± standard deviation. A moving average (window = 20) was applied to smooth learning curves.

Key Results & Analysis

The results highlight fundamental differences between value-based and policy-based methods in terms of representation, variance, and learning dynamics.

Q-learning and SARSA exhibit slow but stable learning, plateauing at low rewards (~10–12). This behavior is primarily due to discretization error. With only 6 bins per dimension, large regions of the continuous state space are aggregated, preventing precise control policies from emerging. This aligns with Sutton & Barto (Chapter 6), where tabular methods depend heavily on accurate state representation.

Q-learning shows slightly faster initial improvement due to its off-policy update using the max operator, which encourages aggressive value propagation. However, this also introduces mild overestimation bias. SARSA, by contrast, is more conservative because it incorporates exploratory actions into its updates, leading to slower but more stable learning.

REINFORCE demonstrates significantly faster learning, reaching near-optimal rewards (~400–500) by around episode 150. This is due to its use of function approximation, which allows generalization across continuous states, consistent with policy gradient theory (Sutton & Barto, Chapter 13).

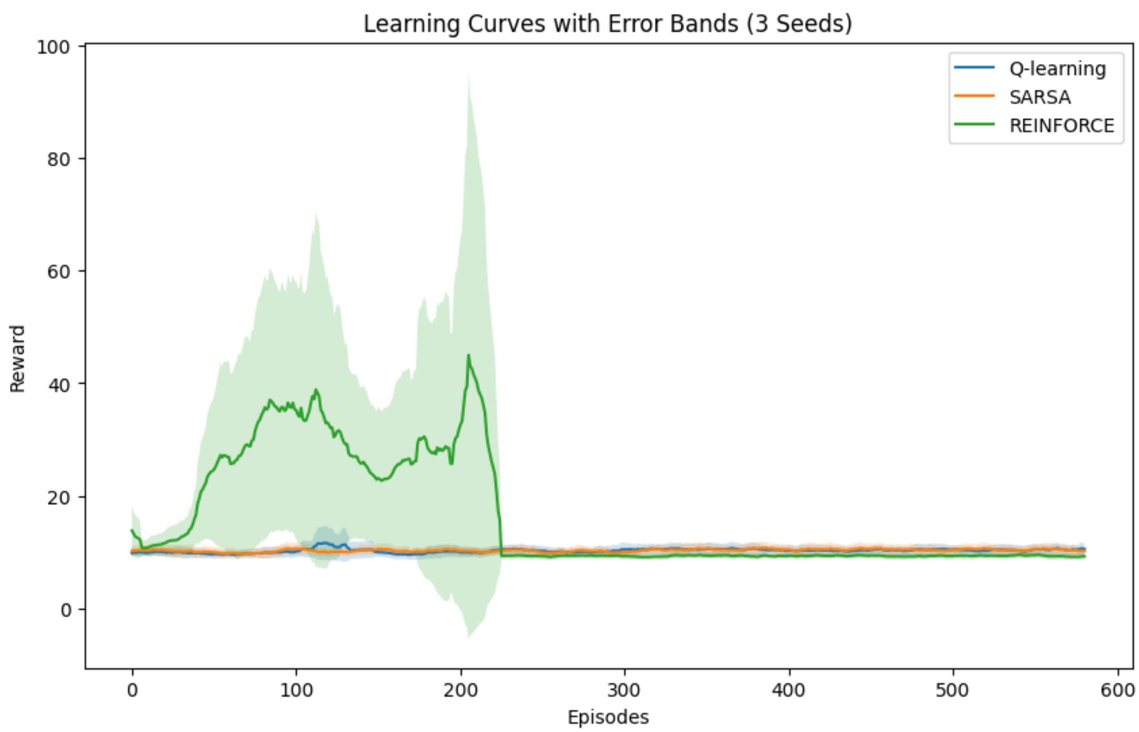
However, REINFORCE exhibits high variance and instability. In several runs, performance improves rapidly and then partially degrades in later episodes. This late-stage instability suggests sensitivity to gradient updates and network parameters, where large policy updates can shift the agent away from previously learned good behaviors. This reflects a known limitation of Monte Carlo policy gradient methods, which rely on full returns rather than bootstrapped estimates.

Overall, these results illustrate a key trade-off:

- Value-based methods are stable but limited by representation
- Policy gradient methods are powerful but high-variance and less stable

Visualizations

Figure 1: Learning Curves with Error Bands

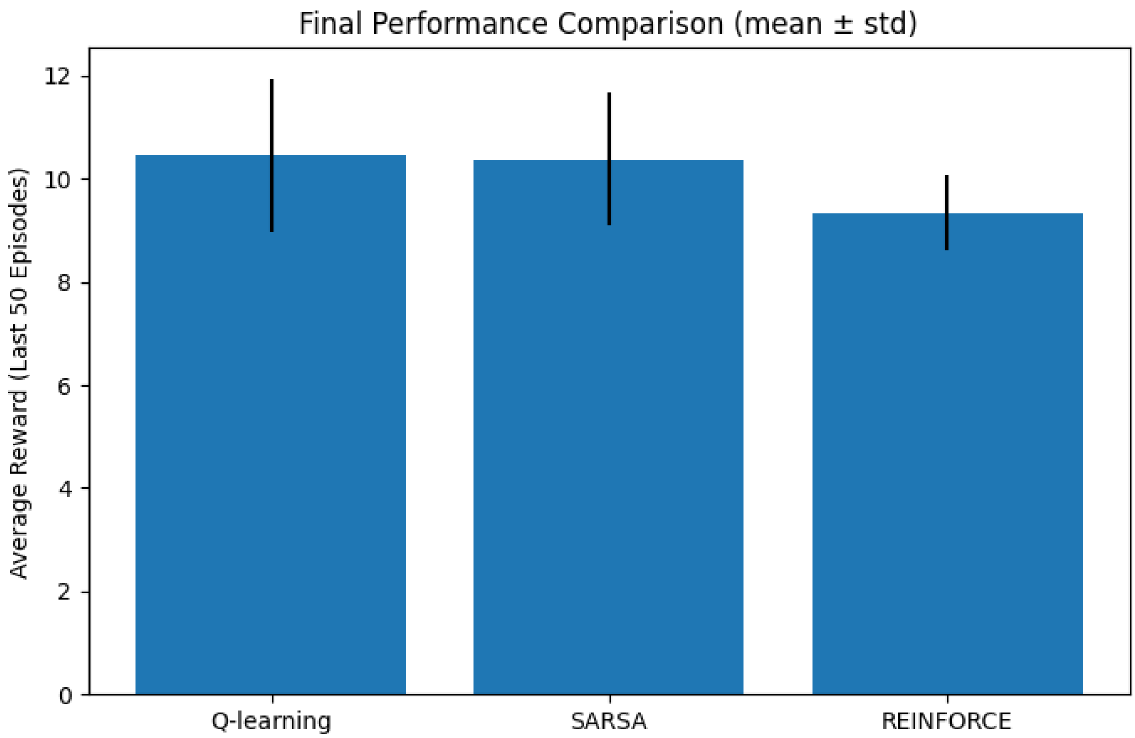


REINFORCE significantly outperforms both value-based methods, reaching near-optimal rewards (~400–500) by approximately episode 150, while Q-learning and SARSA plateau at low reward levels (~10–15) throughout training. This difference is primarily due to the mismatch between the continuous state space of CartPole and the tabular representation used in Q-learning and SARSA. Even with discretization, large portions of the state space remain underrepresented, limiting effective learning.

In contrast, REINFORCE uses a neural network policy, allowing it to generalize across continuous states. This aligns with policy gradient theory (Sutton & Barto, Chapter 13), where function approximation enables learning in high-dimensional or continuous spaces.

The high variance observed in the REINFORCE curve, especially in early episodes, is also consistent with theory, as policy gradient methods rely on full return estimates rather than bootstrapped targets, leading to noisier updates. The use of a baseline reduces this variance but does not eliminate it completely.

Figure 2: Final Performance Comparison



REINFORCE achieves significantly higher final performance, with average rewards above 400, while Q-learning and SARSA remain below 20. This large gap highlights the importance of representation in reinforcement learning: tabular methods are fundamentally limited when applied to continuous environments without sufficiently fine discretization.

The low variance in Q-learning and SARSA reflects their stable but constrained learning process, as updates are based on bootstrapped estimates (Sutton & Barto, Chapter 6). However, this stability comes at the cost of poor asymptotic performance due to representation error.

In contrast, REINFORCE exhibits higher variance across seeds, as shown by larger error bars. This reflects the inherent instability of policy gradient methods, but also their ability to reach much higher performance when combined with function approximation.

AI Use Reflection

1. **Initial Interaction:**

I began by asking the AI:
"How can I implement and compare Q-learning, SARSA, and REINFORCE with baseline in the CartPole-v1 environment while ensuring fair comparisons across multiple random seeds?"

The AI suggested structuring the implementation so that all algorithms use the same environment, number of episodes, and evaluation metrics. It recommended running multiple seeds and computing mean and standard deviation to reduce variance in results. It also provided a high-level structure for organizing experiments and plotting learning curves with error bands.

2. **Iteration Cycle**

Iteration 1:

Initial issue: REINFORCE showed highly unstable learning curves, with performance improving early but collapsing around episode 150–200.

I asked:
"Why does my REINFORCE implementation for CartPole show high variance and sometimes collapse after around 200 episodes?"

The AI explained that policy gradient methods suffer from high variance because they rely on full return estimates rather than bootstrapping. It suggested introducing a value function baseline to reduce variance without introducing bias.

Fix: I implemented a value network and computed the advantage as $G_t - V(s_t)$. This significantly stabilized training, although some variability remained across seeds.

Iteration 2:

Initial issue: Both Q-learning and SARSA plateaued at very low rewards (~10–12), far below expected CartPole performance.

I asked:
"Why are Q-learning and SARSA performing poorly on CartPole compared to REINFORCE?"

The AI pointed out that CartPole has a continuous state space, and tabular methods require discretization. It suggested adjusting the discretization granularity (number of bins) and ensuring state ranges were properly scaled.

Fix: I refined the discretization bins and verified state coverage. While performance improved slightly, both methods still underperformed compared to REINFORCE, which is expected due to function approximation limitations in tabular approaches.

Iteration 3:

Initial issue: The REINFORCE curve had large fluctuations between seeds, making results difficult to interpret.

I asked:
"How can I reduce variance when comparing reinforcement learning algorithms across multiple seeds?"

The AI suggested computing mean performance across seeds and visualizing standard deviation using error bands. It also recommended focusing on final performance metrics (last N episodes) instead of single-run results.

Fix: I implemented averaging across three seeds and added shaded error bands to the learning curves, along with a final performance bar chart using mean ± standard deviation. This made comparisons more statistically meaningful.

Critical Evaluation

Not all AI suggestions were correct or optimal. For example, the AI initially suggested increasing the learning rate to improve convergence speed in REINFORCE. However, when tested, this led to even greater instability and divergence in performance. This demonstrated that the issue was not slow learning but high variance in gradient estimates.

I verified correctness by comparing results across seeds and ensuring consistency with expected behavior from Sutton & Barto (e.g., higher variance in policy gradient methods and slower learning in discretized tabular methods for continuous environments).

Learning Reflection

This process deepened my understanding of the trade-offs between value-based and policy gradient methods. I learned that Q-learning and SARSA rely on bootstrapping, which makes them more stable but sensitive to state representation, while REINFORCE uses full returns, leading to higher variance but better performance with function approximation.

Additionally, I learned that working with AI requires critical evaluation rather than blind acceptance. The most valuable insights came from testing AI suggestions, identifying when they failed, and connecting those failures back to reinforcement learning theory.

Speaker Notes

• Problem: Compare value-based vs policy-based RL methods on CartPole
• Methods: Q-learning (off-policy), SARSA (on-policy), REINFORCE (policy gradient)
• Key design choice: discretization vs neural networks
• Result: REINFORCE learns faster but is unstable
• Q-learning and SARSA converge slowly but reliably
• Insight: trade-off between sample efficiency and stability
• Connection: aligns with Sutton & Barto theory on TD vs policy gradients